
1. Blob Count

Program Name: BlobCount.java

Input File: blobcount.dat

Judges Input File

```
5
7 8 2
...*...
****...
****...
****...
...**...
..**...
..**...
..**...
2 1
5 3
5 6 3
...***
***...
...***
**...***
**...***
3 4
4 1
1 4
3 10 5
***.***.*
***.***.
***.*...***
1 1
1 10
1 6
3 5
3 8
7 13 7
*.***.*.***
*.***.*.***
..***.***
..***.***
..***.***
..***.***
..***.***
*.***.*.***
1 1
1 3
1 10
3 1
7 1
4 7
7 11
3 3 1
***
***
***
1 1
```

Judges Output to Screen (see next page)

1. Blob Count (cont.)

Judges Output to Screen

12
NO BLOB
9
4
3
9
1
4
1
3
2
21
12
NO BLOB
1
12
3
9

2. Clock in the Mirror

Program Name: Clock.java Input File: clock.dat

Judges Input File

12
3:48
8:27
10:30
1:23
5:00
12:18
11:02
1:00
6:58
7:07
4:44
12:00

Judges Output to Screen

8:12
3:33
1:30
10:37
7:00
11:42
12:58
11:00
5:02
4:53
7:16
12:00

3. Digi-count

Program Name: Digicount.java

Input File: digicount.dat

Judges Input File

4 12 35 19 71 100 1 51

Judges Output to Screen

0 1 1 1 1 0 0 0 0 0
1 5 2 1 1 1 1 1 1 1
3 14 14 10 4 4 3 3 3 3
1 12 2 2 2 2 2 2 2 2
7 18 17 17 17 17 17 9 7 7
11 21 20 20 20 20 20 20 20 20
0 1 0 0 0 0 0 0 0 0
5 16 15 15 15 7 5 5 5 5

4. Jack Be Nimble

Program Name: Jack.java Input File: jack.dat

Judges Input File

```
JACK BE NIMBLE.  
JACK BE QUICK.  
JACK JUMP OVER  
THE CANDLESTICK.  
JACK FELL ON HIS BUM.  
GOT A KNOT ON HIS HEAD.  
NEEDED MEDS FOR HIS TUM  
AND TOOK HIS BOD TO BED.  
1  
2  
3  
4  
5  
6  
7
```

Judges Output to Screen

```
JACK BE QUICK.  
THE CANDLESTICK.  
GOT A KNOT ON HIS HEAD.  
AND TOOK HIS BOD TO BED.  
2  
4  
6
```

5. Maze Solver

Program Name: Maze.java

Input File: Maze.dat

Judges Input File

6

```
#S#####  
#.....####.....####  
####.#####  
####.#####.####  
####.#####.####  
####.#####.####  
#####.#####.####  
#####.###.#####  
#.....####.....####  
##.##.#####  
##.##.....#####  
##.#####.#####  
##....#####.##...##  
#####.##.##.#####  
#####.####...#####  
#####.####.#####  
##.....##  
#####.##  
###.....##  
###E#####
```

```
#####  
#.....####.....####  
####S.#####  
####.#####.####  
####.#####.####  
####.#####.####  
#####.#####.####  
#####.####.....####  
#.....###.....##  
#####.#####  
#####.....#####  
##.....####.#####  
#####.#####  
#####.##.##...##  
#####.#####  
#####.#####  
##.....##  
#####  
###.....##  
###E#####
```

(continued on next page)

5. Maze Solver (cont.)

```
#####
#####
####.#####
#####
####.#####.####
####...#####.####
#####.#####...###
#####.#####
#.....#####
#####.#####
#####.....#####
#####.#####
#####.#####
#####.###.#####
#####.#####
#####S#####
#####
###.....#####.##
#####
###E#####
```

```
#####
#.....#
####.###.....####
#####.....####
####.####.###.####
####...#####.####
#####.#####...###
#####.#####.#####
#.....####.....###
#####.....###
#####.....#####
#####.#####
##...#####.##.##
###.###.##.#####
#####.##...##
##.....S.....###
#####.#####
#####.#####.##
#####.....#####
###E###...#####
```

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5. Maze Solver (cont.)

```
#####
#.....####.....####
####..#####
####.#####.####
####.#####.####
####...#####.####
#####.#####...####
#####..#####
#.....####SE#####
#####.#####
#####.....#####
#####.#####
#####.#####
#####..##.#####
#####..#####
#####.#####
#####.#####
##.....##
#####.##
###.....##
#####
```

```
#####
#####
#####SE#####
#####
#####
#####
#####
#####
#####
#####
#####
#####
#####
#####
#####
#####
#####
#####
#####
#####
```

Judges Output to Screen

```
Maze #1: YES
Maze #2: NO
Maze #3: NO
Maze #4: NO
Maze #5: YES
Maze #6: YES
```

6. Octagon Means Stop

Program Name: Octagon.java

Input File: octagon.dat

Judges Input File

6 7 5 15 10

Judges Output to Screen

```
xxxxxxx
  x      x
 x      x
x      x
x      x
x      x
x      x
x      x
x      x
x      x
x      x
xxxxxxx

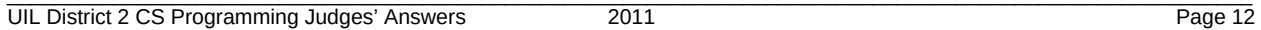
xxxxxxxxx
  x      x
 x      x
x      x
x      x
x      x
x      x
x      x
x      x
x      x
x      x
x      x
x      x
x      x
x      x
xxxxxxx

xxxxxxx
  x      x
 x      x
x      x
x      x
x      x
x      x
x      x
x      x
x      x
x      x
xxxxxxx
```

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[illegible]

UIL District 2 CS Programming Judges' Answers



7. Pal Palindromes

Program Name: Pals.java Input File: pals.dat

Judges Input File

```
12
5 2
7 3
20 5
12 4
8 2
15 4
25 3
13 5
50 7
49 3
19 2
19 7
```

Judges Output to Screen

```
1
12
0
7
1
3
2
69
0
2
0
471
```

8. Post Office

Program Name: PostOffice.java

Input File: postoffice.dat

Judges Input File

```
11
1.21 0.33 1.42 0.90 0.45 1.30 0.30
1.20 0.59 1.23 0.99
0.62 0.61 0.63 0.64 0.65 0.66 0.67 0.68 0.69
2.35
4.60
4.59
1.20 1.20 1.20 1.20
1.25 1.25 1.25
2.22 1.03 1.10 0.55 0.34 0.99
1.12 1.23 1.20 0.55 0.34
0.22 0.22 0.22 0.12 0.22 0.22 0.32 0.22 0.22 0.52 0.22 0.52 0.22 0.22 0.22
0.22
```

(The eleventh line of the actual input file appears to span two lines in the printed version.)

Judges Output to Screen

```
OVERWEIGHT
OK
OVERWEIGHT
OK
OVERWEIGHT
OK
OVERWEIGHT
OK
OVERWEIGHT
OK
OK
```

9. Rattle

Program Name: Rattle.java **Input File:** rattle.dat

Judges Input File

7

```
N EOAIULEVESCIMPCT*R COE
EIG IIKTWINTIIG UHL DISSR CT PSTTWERF RTTTS*EB DOFAHDJSLAIENANC
CEELGYIIIA A ROA LO BBGLF N DFYFAOTDAFLHTO AM*AB
AOCDE TEUERNWVRRSSWORE *EBED FPRKEN
AELFIOLBVL !O TO
LP*MYN000
WSRIDNAECCMAUHES ACEETCE GOECTES RE PHE IR AHE TOPLO*LB ONPTTER
```

Note: There is an intentional space at the end of data cases 2 and 4

Judges Output to Screen

```
I LOVE COMPUTER SCIENCE
WINNING UIL DISTRICT IS THE FIRST STEP TOWARD STATE
I AM A REALLY BIG FAN OF FOOTBALL TODAY
TRUER WORDS WERE NEVER SPOKEN
I LOVE FOOTBALL!
MONOPOLY
COMPUTER SCIENCE COACHES ARE THE GREATEST PEOPLE IN THE WORLD
```

10. Spring Trip

Program Name: Spring.java

Input File: spring.dat

Judges Input File

```
9
STEVE 4
T3 P5 D100
T4 D25
P3 T4
D100
MARY 3
D300 T3 P5
T8 P8
P10
JOE 5
P20 D90 T25
D24 P17 D75
D25
P20
D100
ANDREA 3
T4 P2 D75
P5 D100 T8
D75 P12 T13
GEORGE 5
T12
D195
P13
D105 T10 P12
D79 T11
ALEX 0
ROCKY 1
D500
RODNEY 2
T14
P20
ANDREW 1
D499
```

Judges Output to Screen

```
STEVE $146.99
MARY OVER $28.81
JOE OVER $365.79
ANDREA OVER $46.43
GEORGE OVER $269.75
ALEX $500.00
ROCKY $0.00
RODNEY $271.10
ANDREW $1.00
```

11. Values

Program Name: Values.java

Input File: values.dat

Judges Input File

```
15
HARD WORK
QUIT
SCHOOL
KNOWLEDGE IS POWER
HOMEWORK
ABC
STATE UIL
COMPUTER SCIENCE
GEOMETRY IS FUN
ARITHMETIC A
MATHEMATICS
TRIGONOMETRY
UIL STATE
BOARDWALK
USE A TILT
```

Judges Output to Screen

```
6 ABC
67 QUIT
72 SCHOOL
87 BOARDWALK
98 HARD WORK
107 ARITHMETIC A
107 STATE UIL
107 UIL STATE
107 USE A TILT
108 HOMEWORK
112 MATHEMATICS
169 COMPUTER SCIENCE
177 GEOMETRY IS FUN
179 TRIGONOMETRY
201 KNOWLEDGE IS POWER
```

12. XML Checker

Program Name: XML.java

Input File: xml.dat

Judges Input File

```
5
4
<student>
<name>
John Doe
</student>
2
<gpa>
</gpa>
8
<address>
<street>
123 Any Street
</street>
<city>
Any Town
</city>
</address>
8
<address>
<street>
123 Any Street
</street>
<city>
Any Town
</address>
</city>
16
<book>
<title>
Alice's Adventures in Wonderland
</title>
<author>
<first_name>
Lewis
</first_name>
<last_name>
Carroll
</last_name>
</author>
<publisher>
Tribeca Books
</publisher>
</book>
```

Judges Output to Screen

```
invalid
valid
valid
invalid
valid
```